

EJERCICIOS CON GPG

GPG (o **GnuPG**) es una herramienta de **cifrado y firma digital** muy usada en Linux/Windows/macOS. Sirve para:

¿Cómo funciona a nivel simple?

GPG trabaja principalmente con **dos claves**:

-  **Clave privada**: solo la tienes tú (NUNCA se comparte). Sirve para descifrar y firmar.
-  **Clave pública**: se puede compartir. Sirve para cifrar para ti y verificar tus firmas.



Ejemplo típico:

- Yo quiero mandarte un archivo secreto → lo cifro con **tu clave pública**
- Solo tú podrás descifrarlo con **tu clave privada**

◆ Generación y gestión de claves

1. **Comprobar versión de GPG**
2. `gpg --version`
3. **Crear un par de claves RSA nuevo**
4. `gpg --full-generate-key`

(usar nombre ficticio y correo de práctica)

```
gpg: /home/jcsantos/.gnupg/trustdb.gpg: trustdb created
gpg: directory '/home/jcsantos/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/jcsantos/.gnupg/openpgp-revocs.d/9264CDEEE6CE1D42C00E89BC329913DAF533502A.rev'
public and secret key created and signed.

pub   rsa3072 2026-01-26 [SC]
      9264CDEEE6CE1D42C00E89BC329913DAF533502A
uid           Parrot <flumperio@gmail.com>
sub   rsa3072 2026-01-26 [E]

jcsantos at parrot in ~
```

5. **Listar claves públicas propias**

6. `gpg --list-keys`

```
jcsantos at parrot in ~
└─○ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid: 1  signed: 0  trust: 0-, 0q, 0n, 0m, 0f, 1u
/home/jcsantos/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-01-26 [SC]
      9264CDEEE6CE1D42C00E89BC329913DAF533502A
uid           [ultimate] Parrot <flumperio@gmail.com>
sub   rsa3072 2026-01-26 [E]
```

7. Listar claves privadas (secretas)

8. `gpg --list-secret-keys`

```
jcsantos at parrot in ~
└─○ gpg --list-secret-keys
/home/jcsantos/.gnupg/pubring.kbx
-----
sec   rsa3072 2026-01-26 [SC]
      9264CDEEE6CE1D42C00E89BC329913DAF533502A
uid   [ultimate] Parrot <flumperio@gmail.com>
ssb   rsa3072 2026-01-26 [E]

jcsantos at parrot in ~
└─○
```

9. Exportar clave pública a un archivo

10. `gpg --armor --export "NOMBRE" > clave_publica.asc`

```
jcsantos at parrot in ~
└─○ gpg --armor --export "NOMBRE" > clave_publica.asc
gpg: WARNING: nothing exported
jcsantos at parrot in ~
└─○ gpg --armor --export "Parrot" > clave_publica.asc
jcsantos at parrot in ~
└─○ cat clave_publica.asc
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQGNBGL3V14BDADJK6n5ZcCc0eUIQ1rF9ansYUpLWCXx96Yjy7o5Iu7dc5meSsE+
yKV9LZwKJxmPTcZ0JEKBSjLHgaC5WcKvc5lUA2wLU3qBiKWZD8ZgG9Ar0HewPSFB
TJ+ugo7S/sTQLwYo83ajYHwvVG2ez0TnfzF5zMUE6dzuEbPZk47vuaJxchSeeQ3D
3tLz7av7I+O8tbhyAeoP7c+/G2ngphvzW/lq3jL/KHwmfWoh0XQlnIfg4v+1ZT1S
nQaSUxKC9hDmcmKY584pt5U16lSK/vch60kCbg063l9ZGPt773biHgigfhexouNe
1ZFd6/M8oIZB83+59CSKuFMq5Lr87Vfhymw1TMch5BvmtZLDJ9FknFTSa8nDj73v
O2aXKPLN6Pm6KuKiE/bHiZhbYaOT72GIMhY03YVLD4JtGb2DTuz7BQcu4dGuuK43
pP+i5jTuaVDedyCTDVe3wA13x70UT/Q4Gc0Gf+0510x1e7eHicTkmhFDjvbgRmy6
-----
```

11. Exportar clave privada (para backup)

12. `gpg --armor --export-secret-keys "NOMBRE" > clave_privada.asc`

```
jcsantos at parrot in ~
└─○ gpg --armor --export-secret-key "Parrot" > clave_privada.asc
jcsantos at parrot in ~
└─○ cat clave_privada.asc
-----BEGIN PGP PRIVATE KEY BLOCK-----

lQWGBGL3V14BDADJK6n5ZcCc0eUIQ1rF9ansYUpLWCXx96Yjy7o5Iu7dc5meSsE+
yKV9LZwKJxmPTcZ0JEKBSjLHgaC5WcKvc5lUA2wLU3qBiKWZD8ZgG9Ar0HewPSFB
TJ+ugo7S/sTQLwYo83ajYHwvVG2ez0TnfzF5zMUE6dzuEbPZk47vuaJxchSeeQ3D
3tLz7av7I+O8tbhyAeoP7c+/G2ngphvzW/lq3jL/KHwmfWoh0XQlnIfg4v+1ZT1S
nQaSUxKC9hDmcmKY584pt5U16lSK/vch60kCbg063l9ZGPt773biHgigfhexouNe
1ZFd6/M8oIZB83+59CSKuFMq5Lr87Vfhymw1TMch5BvmtZLDJ9FknFTSa8nDj73v
-----

jcsantos at parrot in ~
└─○ diff clave_privada.asc clave_publica.asc
1c1
< -----BEGIN PGP PRIVATE KEY BLOCK-----
-
-
-
> -----BEGIN PGP PUBLIC KEY BLOCK-----
3c3
< lQWGBGL3V14BDADJK6n5ZcCc0eUIQ1rF9ansYUpLWCXx96Yjy7o5Iu7dc5meSsE+
-
-
> mQGNBGL3V14BDADJK6n5ZcCc0eUIQ1rF9ansYUpLWCXx96Yjy7o5Iu7dc5meSsE+
11,83c11,41
< 5tPS/0wzo4CzjjUAQEAAf4HAWKGeOs0jnrQNV+iPAN4h8I3KwVi/kmLWpmDZdo5
< EjMysYNFP3DnR7ocd1USiqPKaZBbyM1KtVDeqER+kNDxjB6mVmK5pnU2B1LQkhXZ
< S0sbNgateobYtxPuRgmpV5LZuvky9u7SETP/kvYm8VRDrP3WVjekdEke+OKjf0Jp
< rXbEy/qUEh0S7vc0ikY8/Vkdtawn0Iu0Z2dyKJ31eUmTsA02dHHgeqf8W1lBtRwB
< J9b8dKXAcuUAggDLlzS4Dx/ZYticeX0Gm4RIjoDp6PA+CCcDXz1934FZs3wbXfUg
< 4b8dPtiHfA3Yv573UT/0p6GdTiHvDp6G88YQTIvUuLc7z0p6G4u4FUPF14
-----
```

◆ Cifrado y descifrado

7. **Crear un archivo de texto simple**
8. `echo "Hola, esto es un mensaje secreto" > mensaje.txt`
9. **Cifrar con la clave pública propia**
10. `gpg -e -r "NOMBRE" mensaje.txt`
11. **Ver el archivo cifrado (mensaje.txt.gpg)**
12. `cat mensaje.txt.gpg`

```
jcsantos at parrot in ~/ejercicios/gpg
└─ cat fichero_cifrado_gpg.txt
Hola. Mi jaca galopa y corta el viento cuando pasa por el puerto caminito de Jerez.
```

```
jcsantos at parrot in ~/ejercicios/gpg
└─ gpg -e -r "Parrot" fichero_cifrado_gpg.txt
```

```
jcsantos at parrot in ~/ejercicios/gpg
└─ cat fichero_cifrado_gpg.txt
Hola. Mi jaca galopa y corta el viento cuando pasa por el puerto caminito de Jerez.
```

```
1. ┌─ jcsantos at parrot in ~/ejercicios/gpg
   │└─ cat fichero_cifrado_gpg.txt.gpg
   └─ @@@0@)??
        @@@@@qQ, Z@@9@-?          ?I5@@@@?)@<DT.@x??      ?/?!@)iq##?mbB@BiZHLqb?-e???e????#@Y(???2)    ?'.?-?'@-??zP?.z?Z?"
u.%pl??(????bz^A?u?:b?H6'#####[?]@)?i?#'*8@{E@@@@b?D@[QQ
                                     ^H?3;-?@DV?7????###Ax'Q####Z@?)e??2???3I@a=d{@@g??3???n?@?, ??
26 I?@??_ z:[???@oby
26                               @@S Y(????)@@@@jyF?(qnD?`?4?*Fn'?@i)?[Q$ i#@@@@ue@@?)??7/QYKdX10@;.D^<?@e??*os^@z@
26                                ??????/))@@@@T6a@po
r??-_???
```

```
jcsantos at parrot in ~/ejercicios/gpg
```

- 13. Descifrar con la clave privada**
- ```
gpg -d mensaje.txt.gpg
```
- 11. Guardar el descifrado en archivo**
- ```
gpg -d mensaje.txt.gpg > mensaje_descifrado.txt
```

```
jcsantos at parrot in ~/ejercicios/gpg
└─ ll
total 20K
drwxr-xr-x 1 jcsantos jcsantos 168 Jan 26 13:17 .
drwxr-xr-x 1 jcsantos jcsantos 48 Jan 26 13:10 ..
-rw-r--r-- 1 jcsantos jcsantos 5.1K Jan 26 13:10 clave_privada.asc
-rw-r--r-- 1 jcsantos jcsantos 2.4K Jan 26 13:10 clave_publica.asc
-rw-r--r-- 1 jcsantos jcsantos 84 Jan 26 13:16 fichero_cifrado_gpg.txt
-rw-r--r-- 1 jcsantos jcsantos 553 Jan 26 13:17 fichero_cifrado_gpg.txt.gpg

jcsantos at parrot in ~/ejercicios/gpg
└─ gpg -d fichero_cifrado_gpg.txt.gpg > fichero_gg_descifrado.txt
gpg: encrypted with 3072-bit RSA key, ID F330F3BF977DF2DE, created 2026-01-26
"Parrot <flumperio@gmail.com>"

jcsantos at parrot in ~/ejercicios/gpg
└─ cat fichero_gg_descifrado.txt
Hola. Mi jaca galopa y corta el viento cuando pasa por el puerto caminito de Jerez.

jcsantos at parrot in ~/ejercicios/gpg
└─
```

◆ Firmas digitales

12. Firmar un archivo en claro

gpg --clearsign mensaje.txt

```
jcsantos at parrot in ~/ejercicios/gpg
└─○ gpg --clearsign fichero_cifrado_gpg.txt
└─○ ll
total 28K
drwxr-xr-x 1 jcsantos jcsantos 272 Jan 26 13:21 .
drwxr-xr-x 1 jcsantos jcsantos 48 Jan 26 13:10 ..
-rw-r--r-- 1 jcsantos jcsantos 5.1K Jan 26 13:10 clave_privada.asc
-rw-r--r-- 1 jcsantos jcsantos 2.4K Jan 26 13:10 clave_publica.asc
-rw-r--r-- 1 jcsantos jcsantos 84 Jan 26 13:16 fichero_cifrado_gpg.txt
-rw-r--r-- 1 jcsantos jcsantos 792 Jan 26 13:21 fichero_cifrado_gpg.txt.asc
-rw-r--r-- 1 jcsantos jcsantos 553 Jan 26 13:17 fichero_cifrado_gpg.txt.gpg
-rw-r--r-- 1 jcsantos jcsantos 84 Jan 26 13:19 fichero_gg_descifrado.txt
└─jcsantos at parrot in ~/ejercicios/gpg
```

13. Generar una firma separada (.sig)

gpg --detach-sign mensaje.txt

14. Verificar una firma

gpg --verify mensaje.txt.sig mensaje.txt

```
File Edit View Search Terminal Help
└─○ gpg --verify fichero_cifrado_gpg.txt.sig fichero_cifrado_gpg.txt
gpg: Signature made Mon Jan 26 13:23:10 2026 CET
gpg: using RSA key 9264CDEEE6CE1D42C00E89BC329913DAF533502A
gpg: Good signature from "Parrot <flumperio@gmail.com>" [ultimate]
└─jcsantos at parrot in ~/ejercicios/gpg
```

```
gpg: directly from the user's keyring: 'Kali-gpg' created
gpg: revocation certificate stored as '/home/jcsantos/.gnupg/openpgp-revocs.d
/4F02D225C4BCE8541315C77E4455A97A4FD782A7.rev'
public and secret key created and signed.

pub   rsa3072 2026-01-26 [SC]
      4F02D225C4BCE8541315C77E4455A97A4FD782A7
uid           Kali-gpg <flumperio@gmail.com>
sub   rsa3072 2026-01-26 [E]

(jcsantos@kali)-[~]
└─$
```

◆ Trabajo con otros compañeros

15. Importar la clave pública de otro alumno

```
gpg --import clave_compañero.asc
```

16. Enviar un mensaje cifrado a su clave

```
gpg -e -r "NombreDelCompañero" mensaje.txt
```

17. Firmar la clave de un compañero

```
gpg --sign-key "NombreDelCompañero"
```

```
jcsantos at parrot in ~/ejercicios/gpg
└─○ gpg --import clave_publica_companiero.asc
gpg: key 4455A97A4FD782A7: public key "Kali-gpg <flumperio@gmail.com>" imported
gpg: Total number processed: 1
gpg:      imported: 1
└─○ nvim mensaje_companiero.txt
jcsantos at parrot in ~/ejercicios/gpg
└─○ gpg -e -r "Kali-gpg" mensaje_companiero.txt
gpg: 36CF94BE8D3D4258: There is no assurance this key belongs to the named user

sub rsa3072/36CF94BE8D3D4258 2026-01-26 Kali-gpg <flumperio@gmail.com>
Primary key fingerprint: 4F02 D225 C4BC E854 1315 C77E 4455 A97A 4FD7 82A7
Subkey fingerprint: BCE7 4AD0 97DA 4619 0BE6 853E 36CF 94BE 8D3D 4258

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
└─○ gpg --sign-key "Kali-gpg"

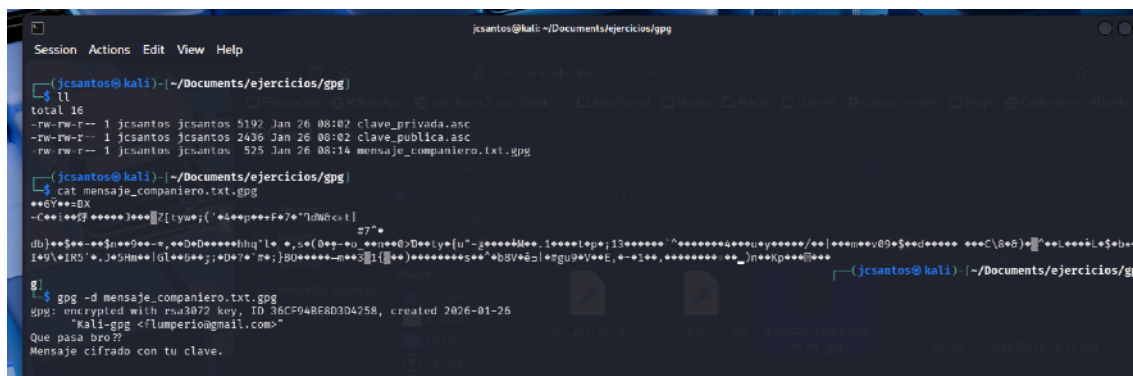
pub rsa3072/4455A97A4FD782A7
created: 2026-01-26 expires: never usage: SC
trust: unknown validity: unknown
sub rsa3072/36CF94BE8D3D4258
created: 2026-01-26 expires: never usage: E
[ unknown] (1). Kali-gpg <flumperio@gmail.com>

pub rsa3072/4455A97A4FD782A7
created: 2026-01-26 expires: never usage: SC
trust: unknown validity: unknown
Primary key fingerprint: 4F02 D225 C4BC E854 1315 C77E 4455 A97A 4FD7 82A7

Kali-gpg <flumperio@gmail.com>

Are you sure that you want to sign this key with your
key "Parrot <flumperio@gmail.com>" (329913DAF533502A)

Really sign? (y/N) y
```



```
jcsantos@kali: ~/Documents/ejercicios/gpg
Session Actions Edit View Help

(jcsantos@kali) ~/Documents/ejercicios/gpg
└─$ ll
total 16
-rw-rw-r-- 1 jcsantos jcsantos 5192 Jan 26 08:02 clave_privada.asc
-rw-rw-r-- 1 jcsantos jcsantos 2436 Jan 26 00:02 clave_publica.asc
-rw-rw-r-- 1 jcsantos jcsantos 525 Jan 26 08:14 mensaje_companiero.txt.gpg
└─(jcsantos@kali) ~/Documents/ejercicios/gpg
└─$ cat mensaje_companiero.txt.gpg
--GPG--BX
--Cee:eej] eeeee]eeee]2[tyu*;*(*eep*eeFe*7e*7dW6c:tl
          37"
db]ee$ee--ee$ee9ee--e,*eDeDeeeee]hqu"le,*s*(Deq-eu,*ee@>Deelye[u"-jceee4Mee,1eeeee]e;13eeeee^eDeee4eeeyeeeee/ee|eeeee;e9e$eeDeeeee eeC\8e6)*eeLeeeLe$ee+
Ie9\*IR5'\*JeeHme*[l*ee6eej;*De7e*ne;]BU+eeee-m+e3]i[ee*)eeeee+e*eb8V+e=|e7gu9e]eeE,*-e]ee,eeeee ee_]ne+Kp+ee]ee
└─(jcsantos@kali) ~/Documents/ejercicios/gp
└─$ gpg -d mensaje_companiero.txt.gpg
gpg: encrypted with rsa3072 key, ID 36CF94BE8D3D4258, created 2026-01-26
"Kali-gpg <flumperio@gmail.com>"
Que pasa bro??
Mensaje cifrado con tu clave.
```

◆ Gestión de claves y revocación

18. Revocar una clave (generar certificado de revocación)

`gpg --output revocacion.asc --gen-revoke "NOMBRE"`

```
jcsantos at parrot in ~/ejercicios/gpg
└─○ gpg --output revocacion.asc --gen-revoke "Parrot"

sec  rsa3072/329913DAF533502A 2026-01-26 Parrot <flumperio@gmail.com>

Create a revocation certificate for this key? (y/N) y
Please select the reason for the revocation:
  0 = No reason specified
  1 = Key has been compromised
  2 = Key is superseded
  3 = Key is no longer used
  Q = Cancel
(Probably you want to select 1 here)
Your decision? 0
Enter an optional description; end it with an empty line:
>
Reason for revocation: No reason specified
(No description given)
Is this okay? (y/N) y
ASCII armored output forced.
Revocation certificate created.

Please move it to a medium which you can hide away; if Mallory gets
access to this certificate he can use it to make your key unusable.
It is smart to print this certificate and store it away, just in case
your media become unreadable. But have some caution: The print system of
your machine might store the data and make it available to others!
jcsantos at parrot in ~/ejercicios/gpg
└─○
```

19. Borrar una clave pública importada

`gpg --delete-key "NombreDelCompañero"`

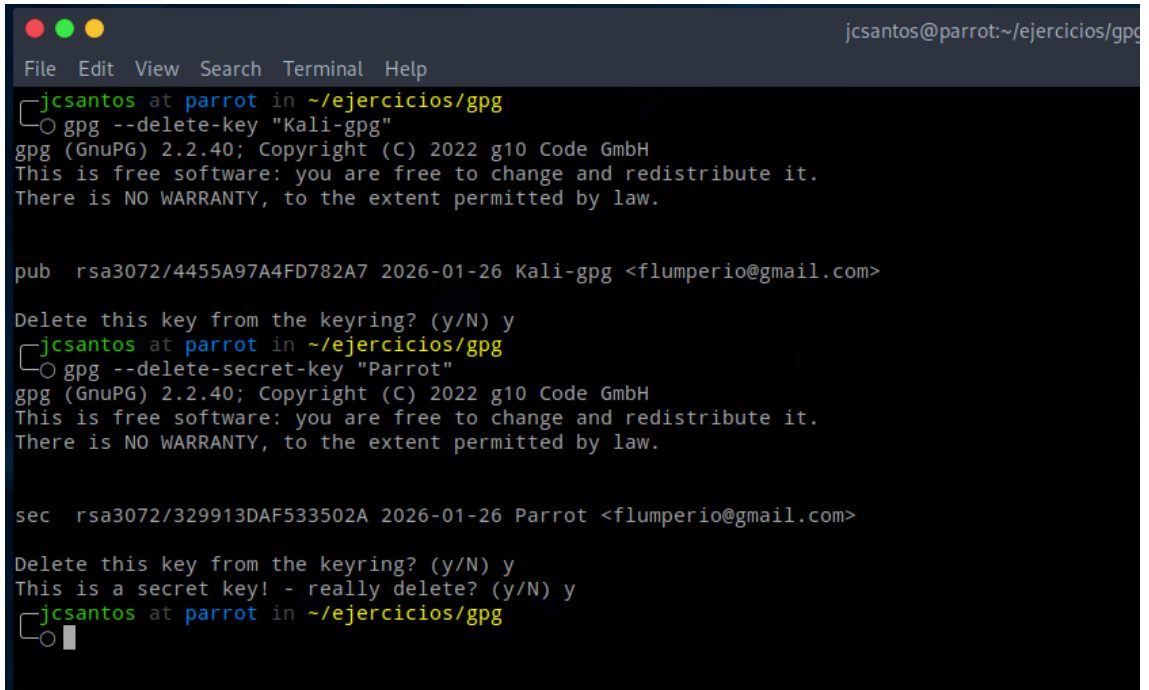
```
jcsantos@parrot:~/ejercicios/gpg
File Edit View Search Terminal Help
└─○ gpg --delete-key "Kali-gpg"
gpg (GnuPG) 2.2.40; Copyright (C) 2022 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

pub  rsa3072/4455A97A4FD782A7 2026-01-26 Kali-gpg <flumperio@gmail.com>

Delete this key from the keyring? (y/N) y
jcsantos at parrot in ~/ejercicios/gpg
└─○
```

20. Borrar tu propia clave (secreta + pública)

`gpg --delete-secret-keys "NOMBRE"`

A terminal window titled 'jcsantos@parrot:~/ejercicios/gpg' with a menu bar (File, Edit, View, Search, Terminal, Help). The user runs 'gpg --delete-key "Kali-gpg"', which shows a public key fingerprint and asks for confirmation to delete it. The user answers 'y'. Then, the user runs 'gpg --delete-secret-key "Parrot"', which shows a secret key fingerprint and asks for confirmation to delete it. The user answers 'y'. The terminal then shows the next prompt.

```
jcsantos at parrot in ~/ejercicios/gpg
└─○ gpg --delete-key "Kali-gpg"
gpg (GnuPG) 2.2.40; Copyright (C) 2022 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

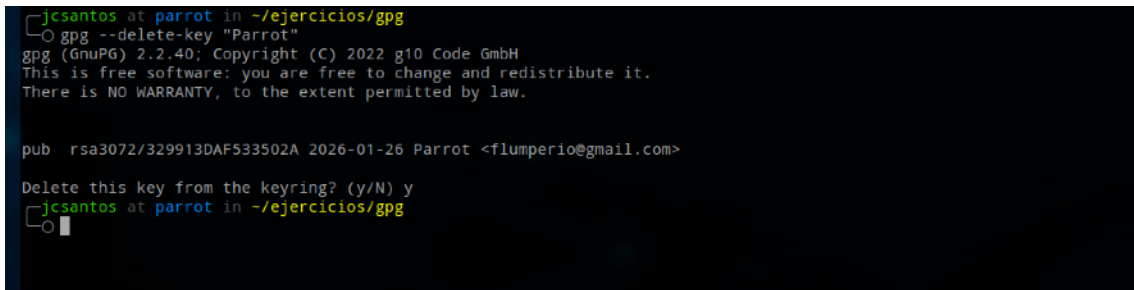
pub  rsa3072/4455A97A4FD782A7 2026-01-26 Kali-gpg <flumperio@gmail.com>

Delete this key from the keyring? (y/N) y
jcsantos at parrot in ~/ejercicios/gpg
└─○ gpg --delete-secret-key "Parrot"
gpg (GnuPG) 2.2.40; Copyright (C) 2022 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

sec  rsa3072/329913DAF533502A 2026-01-26 Parrot <flumperio@gmail.com>

Delete this key from the keyring? (y/N) y
This is a secret key! - really delete? (y/N) y
jcsantos at parrot in ~/ejercicios/gpg
└─○
```

`gpg --delete-key "NOMBRE"`

A terminal window showing the deletion of a public key. The user runs 'gpg --delete-key "Parrot"', which shows a public key fingerprint and asks for confirmation to delete it. The user answers 'y'. The terminal then shows the next prompt.

```
jcsantos at parrot in ~/ejercicios/gpg
└─○ gpg --delete-key "Parrot"
gpg (GnuPG) 2.2.40; Copyright (C) 2022 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

pub  rsa3072/329913DAF533502A 2026-01-26 Parrot <flumperio@gmail.com>

Delete this key from the keyring? (y/N) y
jcsantos at parrot in ~/ejercicios/gpg
└─○
```